

Lecture 3

Labor Markets and the Cost of Inflation

Bank of Portugal—Empirical Macroeconomics

Jonathon Hazell
London School of Economics, Autumn 2025

Today's Plan

- Overall theme: labor markets
- Topics:
 - New costs of inflation from the labor market
 - Wage rigidity and unemployment fluctuations
- New tools:
 - Surveys for eliciting qualitative and quantitative information
 - Data on worker flows, wages, etc

Outline

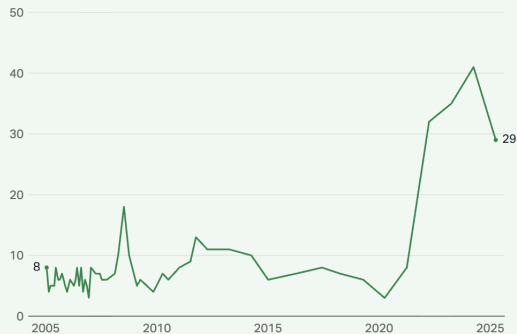
- 1 The Costs of Inflation
- 2 Overview of Search
- 3 Basic Facts
- 4 Basic Search Model
 - Importance of Wage for New Hires
- 5 Empirics of Wage for New Hires
- 6 Costs of Job Loss
- 7 Conclusion

Why Do People Dislike Inflation?

Percentage of Americans Naming Inflation as the Most Important Financial Problem Facing Their Family

What is the most important financial problem facing your family today? [OPEN-ENDED]

— % Mentioning inflation



[Get the data](#) • [Download image](#)

GALLUP

Why Do People Dislike Inflation?

- Classic question, renewed interest
- Lots of disagreement between people and economists!

Costs of Inflation According to Economists

Costs of **steady, expected inflation**:

- **Relative Price Changes:** If all prices and wages do not adjust at the same speed, inflation will change relative prices and lead to inefficient decisions
- **Real Tax Changes:** Many provisions of tax law ignore inflation (e.g., income tax, capital gains tax and interest deductions), could be inefficient
- **Menu Costs:** Firms change price tags/menus more often, pay more attention to prices
- **“Shoe Leather” Costs:** Real value of money falling, people economise on money holdings. Need to make more trips to the ATM/manage between accounts or assets
- **Psychological costs** as in Shiller survey? (certainly more intense bargaining over wage)

Most of the public do not seem to think these costs are important

Costs of Inflation According to Economists

- Volatility?
- Many economists are sceptical that it is possible to have moderate inflation that is stable
- Friedman: “Zero inflation is a politically feasible objective; 10 percent inflation is not. That is the verdict of history.”
- **High inflation has tended to be volatile**
- But is this a law of nature or simply because high inflation periods have been associated with wars and bad economic policies?

Costs of Inflation According to Economists

Costs of **unexpected** inflation:

- Increased Risk and Uncertainty
 - Uncertainty makes financial planning difficult
 - Real value of nominal contracts becomes risky
 - Could write contracts in real terms (index payments to inflation), but costly
 - People are risk averse and dislike uncertainty
- Redistributes wealth arbitrarily
 - Higher unexpected inflation benefits borrowers and harms lenders and vice versa

Costs of Inflation According to the Public

Shiller (1997) survey:

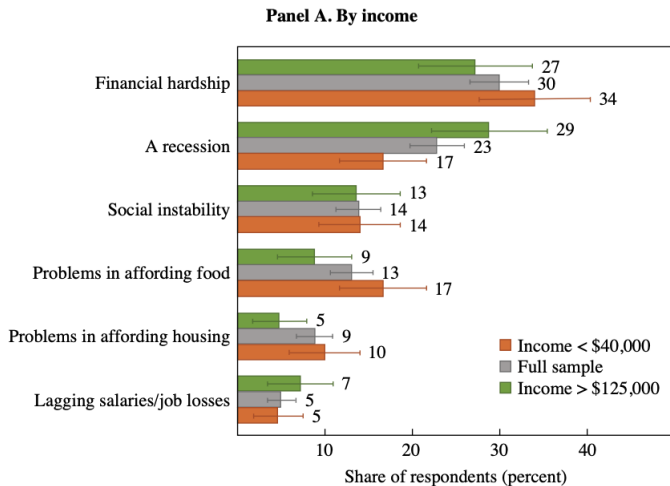
- “biggest gripe about inflation” was “inflation hurts my real buying power; it makes me poorer”
 - 77% of public agrees vs 12% of economists agree (but does depend on source of inflation)
- “Do you agree that preventing high inflation is an important national priority, important as preventing drug abuse?”
 - 52% of public but 18% of economists agree fully
- “I think that if my pay went up, I would feel more satisfaction in my job, more sense of fulfilment, even if prices went up just as much.”
 - 49% of public but 8% of economists agree fully or partially

Costs of Inflation According to the Public

- Stantcheva (2024) updated Shiller by surveying people in the aftermath of the post-Pandemic inflation
- Finds consistent results—people dislike inflation because inflation lowers real buying power, wages do not catch up

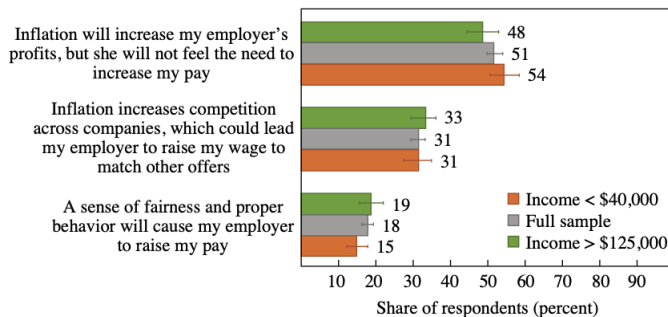
Costs of Inflation According to the Public

Figure 4. If Inflation Increases Too Much, I Worry about . . . [Open-Ended Text]



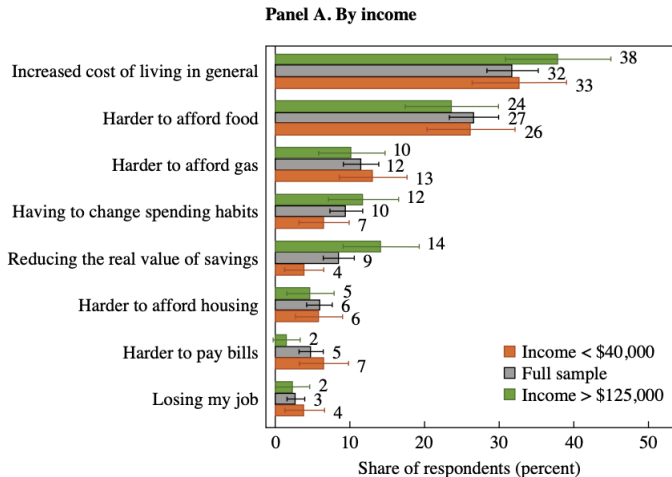
Costs of Inflation According to the Public

Figure 7. Theories about Inflation and Wages



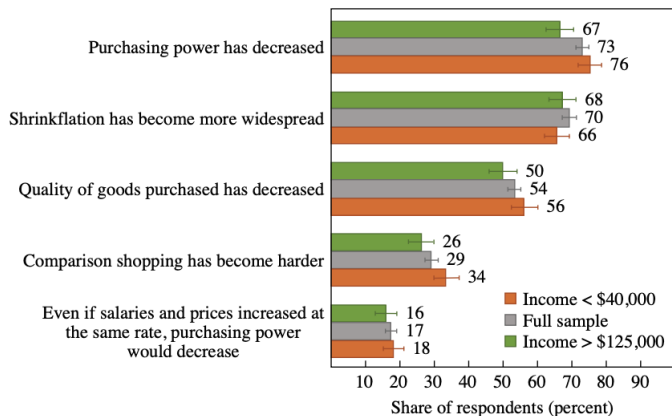
Costs of Inflation According to the Public

Figure 9. The Most Important Impact of Inflation on My Life Has Been . . .
[Open-Ended Text]



Costs of Inflation According to the Public

Figure 10. Inflation Impacts as a Consumer

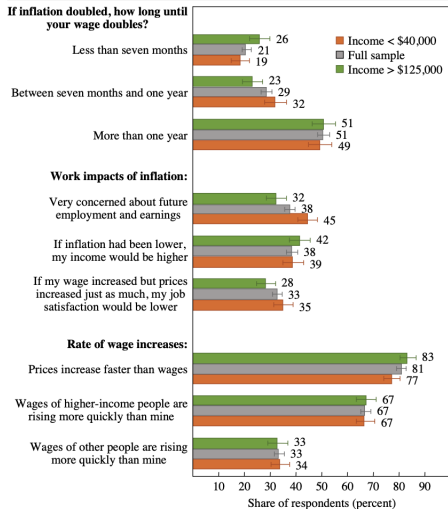


Source: Author's surveys.

Note: The figure reports the share of respondents whose answers are aligned with the statement listed alongside 90 percent confidence intervals. For more details on the questionnaire, see online appendix A.4.

Costs of Inflation According to the Public

Figure 13. Inflation Impacts as a Worker



Costs of Inflation According to the Public

- Summary of the layperson's view:
 - Prices rise, wages don't rise
 - Real wages poorer
- Typical rejoinder by economists (e.g. Mankiw's textbook):
 - Wages **do tend to rise** when prices rise
 - Especially for the case of "pure inflations"
 - Are the public mistaken in their beliefs about the costs of inflation?

Towards New Costs of Inflation

- Needed: a theory of why people dislike inflation
- Captures the fact that people dislike prices rising faster than wages
- Captures that often wages also go up during periods of inflation
- Discipline with data—but how?

Why Do Workers Dislike Inflation?

Wage Erosion and Conflict Costs

Joao Guerreiro¹ Jonathon Hazell² Chen Lian³ Christina Patterson⁴

October, 2025

¹UCLA ²LSE ³UC Berkeley ⁴Chicago Booth

Why do Workers Dislike Inflation?

Why do workers dislike inflation?

- Prior work: **prices rise faster than wages** [Fischer & Modigliani '78; Shiller '97; Stantcheva '24]
- Relevant: how fast nominal wages **keep up with** prices, i.e., how fast **real wages** recover
[Kessel & Alchian '60, Blanco et al '23]

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[Kessel & Alchian '60, Blanco et al '23]

Instead, this paper:

- Workers must take **costly actions** (“**conflict**”) to have wages keep up with prices
- Substantial welfare costs of inflation **even if** real wages do not fall during inflations

1. Motivating survey evidence:

1.1 Workers get wage growth though **costly conflict** w/ firms

[E.g. tough conversations with employers; union strike; soliciting offers from other firms]

1.2 Costly conflict rises with inflation

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2. **Conflict-cost** model: “menu-cost style” model for wage setting
 - Workers consider costly conflict w/ firm to raise nominal wage [+ idiosyncratic shocks]

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2. **Conflict-cost** model: “menu-cost style” model for wage setting
 - Workers consider costly conflict w/ firm to raise nominal wage [+ idiosyncratic shocks]
3. Main analytical result: path of real wages no longer sufficient to inform welfare
 - 3.1 Wage catch-up after inflation from more frequent conflict doesn't raise welfare
 - On the margin: benefit of wage catch-up are offset by cost of conflict [envelope theorem]
 - 3.2 Impact of inflation on worker welfare determined by **wage erosion**
 - How inflation lowers real wages if conflict decision do not change

Quantifying How Conflict Affects the Costs of Inflation

Key term for welfare: size of conflict cost

Survey evidence to quantify *conflict-cost* model:

- Workers willing to sacrifice **1.75% of wages** to avoid conflict

Explore inflation shocks within calibrated model:

→ incorporating conflict **more than doubles** the overall cost of inflation to workers

Inflation and Conflict. [Lorenzoni & Werning '23 a,b]

- **This paper:** related but distinct notion of conflict—costly actions by workers raises wages
- Our focus is the **welfare costs of inflation**, theirs is the causes of inflation

Costs of inflation. [Baily '56; Friedman '77; Fischer & Modigliani '78; Imrohoroglu '92; Lucas '00; Burstein & Hellwig '08; Alvarez et al '16; Nakamura et al '18; Binetti et al '24]

- **Previous:** inflation costs from range of mechanisms
[e.g. shoe leather costs, menu costs, relative price distortions, inflation volatility, tax distortions]
- **This paper:** new + significant costs of inflation via labor markets

Costs of inflation via labor market. [Shiller '97; Hadjini et al '22; Pilossoph & Ryngaert '23; Del Canto et al '23; Ferreira et al '23; Pallotti et al '23; Stantcheva '24; Afrouzi et al. '24; Pilossoph et al '24]

- **This paper:** real wage growth misses costs of inflation in labor market due to conflict

Survey Design

Motivating Evidence: Wage Growth, Inflation and Conflict

The Conflict-Cost Model

Eliciting Conflict Costs in our survey

Quantifying how conflict affects the costs of inflation

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Our Survey

- Survey: **Prolific**
- 3000 prime-age, employed US workers, **Feb-March '24**
- **Representative** of the general population in terms of chosen **observable characteristics**
 - gender, education and political affiliation
- **Survey part I [now]:** Qualitative evidence to motivate model
 - Workers achieve wage growth through costly conflict with employers
- **Survey part II [later]:** Quantitative evidence to discipline the model

Representation

Survey Design

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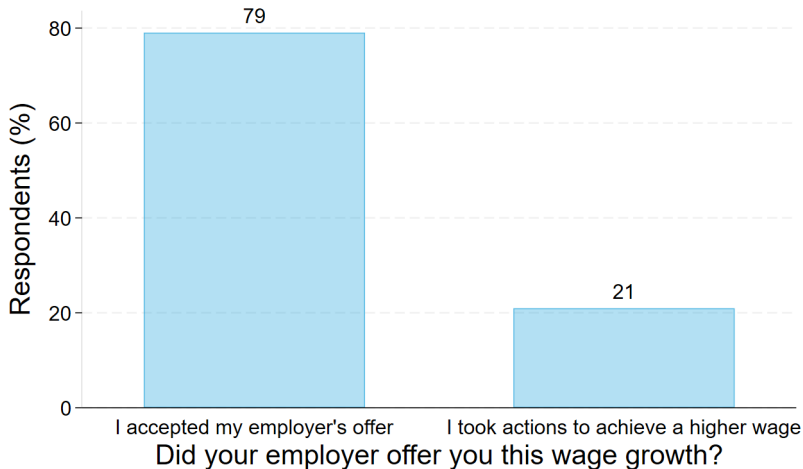
Four findings relating conflict, wage growth, and inflation

1. **Workers choose between accepting employer's default wage offer or conflict**
2. Conflict raises wages
3. Workers who did not conflict believe conflict could have raised wages
4. Inflation leads to conflict: 1 p.p. $\uparrow$ inflation $\Rightarrow$ 1.5 p.p. $\uparrow$ prob. of worker conflict

Workers Choose Employer Offer vs. Conflict to Raise Wage

Heterogeneity

By income



Note: actions include tough conversations with employers; union strike; soliciting other offers, etc.

actions

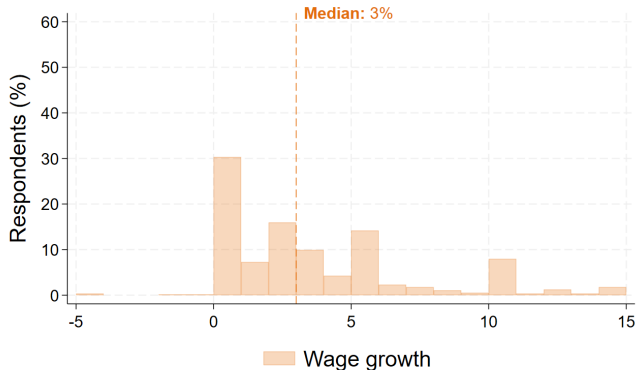
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Conflict Leads to Higher Wages...

Within-individual

High wage workers

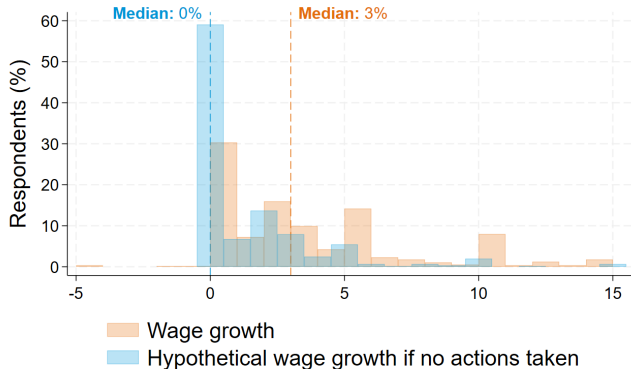


Sample: workers who took costly actions

Conflict Leads to Higher Wages...

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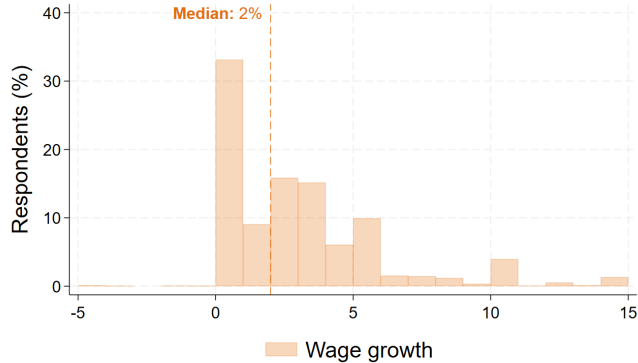


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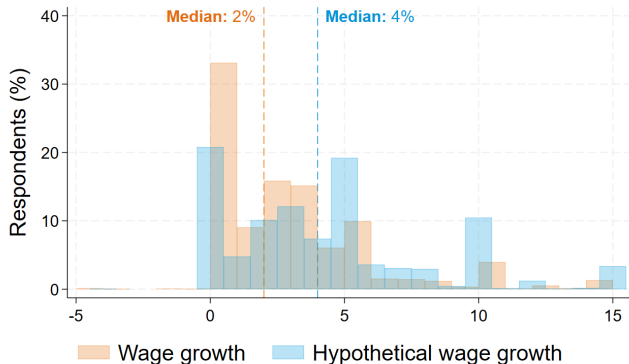
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Workers Sacrifice Wage Growth To Avoid Conflict



Sample: workers who did not take costly actions

Workers Sacrifice Wage Growth To Avoid Conflict



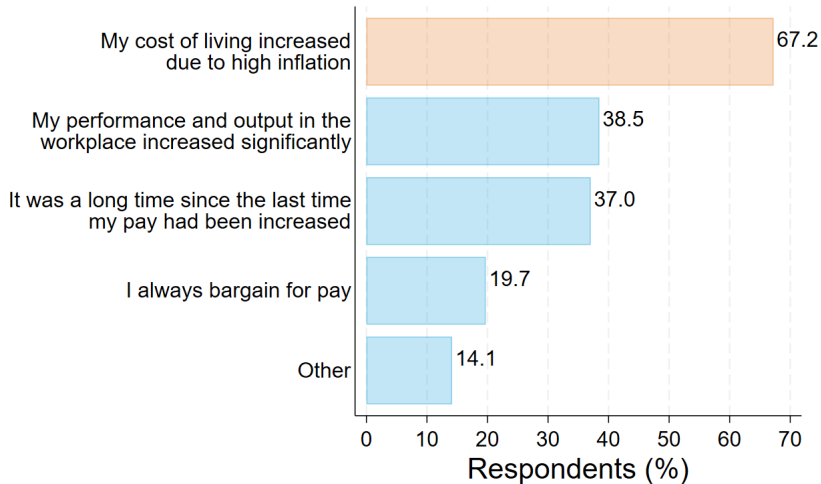
Sample: workers who did not take costly actions

- Workers who do not conflict, report having sacrificed 2pp of wage growth

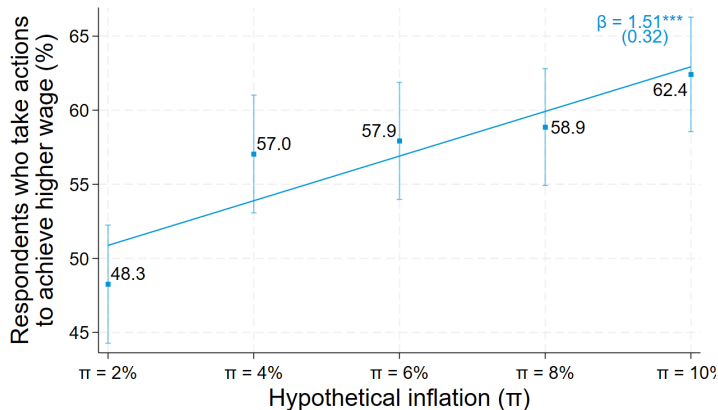
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Inflation was the main motivation for conflict



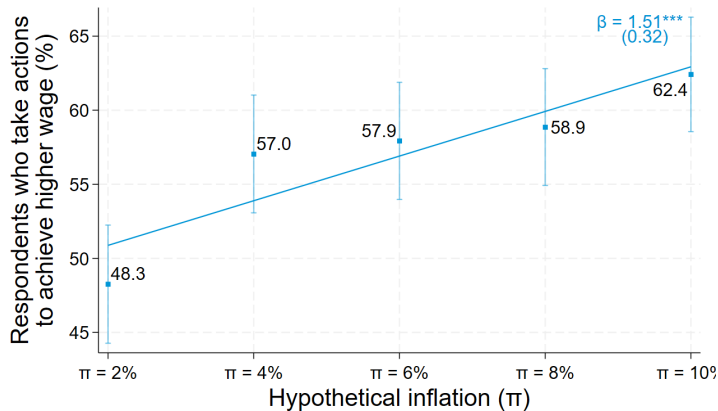
Inflation Increases the Probability of Conflict



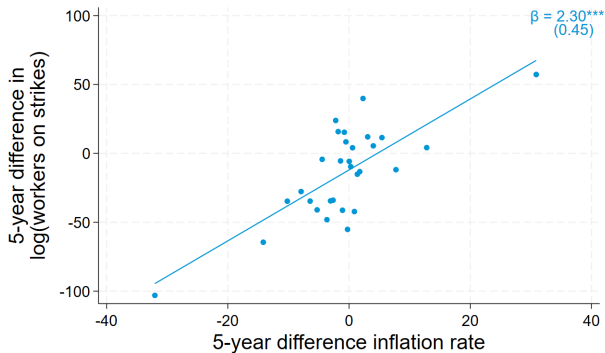
Q: consider a hypothetical situation, inflation is expected to be $x\%$ in next 12 months.

- Where $x \in \{2, 4, 6, 8, 10\}$

Inflation Increases the Probability of Conflict



- 1% $\Delta\pi$ implies a 1.5% increase in conflict – **state-dependent conflict decisions**



$$\Delta \log 100 \times (\text{workers involved in strikes})_{i,t,t-5} = \gamma_i + \lambda_t + \beta \Delta \pi_{i,t,t-5} + \varepsilon_{it},$$

International Labor Organization data, 1960 – 2020

Motivating Evidence: Wage Growth, Inflation and Conflict

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Next: incorporate these features in a “menu-cost style” model of wage setting

[Alvarez et al '16; Nakamura et al. '18; Auclert et al '24]

Survey Design

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Workers' Problem

- A continuum of workers $i \in [0, 1]$ balances **benefit of wage catch up** and **cost of conflict**

$$\max_{\{\mathcal{I}_{i,t}\}_{t=0}^{+\infty}} \mathbb{E}_0 \left[\sum_{t \geq 0} \beta^t (\log c_{i,t} - \kappa_{i,t} \mathcal{I}_{i,t}) \right],$$

where $\mathcal{I}_{i,t} = 1$ if the worker takes costly actions to increase pay and 0 otherwise.

- Now: hand-to-mouth $c_{i,t} = w_{i,t}$. [Paper: borrowing constraints.]
- $\kappa_{i,t}$: i.i.d. “Calvo-plus” cost to increase pay [Nakamura & Steinsson '10; Auclert et al. '23]

$$\kappa_{i,t} = \begin{cases} \kappa & \text{with probability } 1 - \lambda \\ 0 & \text{with probability } \lambda \end{cases}$$

Reduced form cost captures a variety of conflict actions

Workers' Problem

- Real wage

$$w_{i,t} = \begin{cases} w_{i,t-1} e^{\alpha - (1-\gamma)\pi_t} & \text{if } \mathcal{I}_{i,t} = 0 \\ w_{i,t}^* & \text{if } \mathcal{I}_{i,t} = 1 \end{cases}$$

- α : default nominal wage growth without inflation
 - γ : degree of indexation
- Conflict-induced (real) wage $w_{i,t}^*$, keeps up w/ inflation & productivity

$$\log w_{i,t}^* = \log w_{i,t-1}^* + g_z + z_{i,t}$$

$z_{i,t} \in [-\bar{z}, \infty)$ is idiosyncratic productivity shock, g_z is trend productivity growth.

The Impact of Inflation Shocks on Worker Welfare and Wages

- Start from a steady state with inflation $\pi^{ss} \geq 0$.
- An unexpected $t = 0$ **inflation shock**, $\{\hat{\pi}_t\}_{t=0}^{+\infty} = \{\pi_t - \pi^{ss}\}_{t=0}^{+\infty}$. Perfect foresight after.
- Study its impact on aggregate worker welfare

$$\mathcal{W} \equiv \int \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \{ u(c_{i,t}) - \kappa_{i,t} \mathcal{J}_{i,t} \} \right] di$$

and aggregate log real wages

$$\log w_t \equiv \int_0^1 \log(w_{i,t}) di.$$

Model: Higher inflation increases share of conflict

Proposition.

*If $\gamma < 1$, then an **increase in inflation** at $t = 0$ leads to a **larger fraction** of workers engaging in **conflict** at $t = 0$.*

- State dependency: likelihood of conflict increases with inflation
- Unless default wages were fully indexed ($\gamma = 1$)

The Impact of Inflation Shocks on Worker Wages

Definition 1.

The impact of inflation shocks on aggregate log real wage can be decomposed into

$$\hat{w}_t \equiv \log w_t - \log w^{ss} = \hat{w}_t^{\text{erosion}} + \hat{w}_t^{\text{catch-up}},$$

- **Wage erosion**: impact of inflation shocks holding conflict decision at steady state $\mathcal{I}_{i,t}^{ss}$

$$\hat{w}_t^{\text{erosion}} \equiv \omega_t \left(\pi_t, \mathcal{I}_{i,t}^{ss} \right) - \omega_t \left(\pi^{ss}, \mathcal{I}_{i,t}^{ss} \right),$$

where $\omega_t(\pi_t, \mathcal{I}_{i,t})$ is aggregate log real wage at time t given $\pi_t, \mathcal{I}_{i,t}$.

- **Wage catch-up**: impact of inflation shocks through changes in conflict decision $\mathcal{I}_{i,t}$

$$\hat{w}_t^{\text{catch-up}} \equiv \omega_t \left(\pi_t, \mathcal{I}_{i,t} \right) - \omega_t \left(\pi_t, \mathcal{I}_{i,t}^{ss} \right).$$

Theorem.

The (first-order) impact of inflation shocks on worker welfare is given by **wage erosion**

$$\underbrace{\hat{\mathcal{W}}}_{\text{impact on worker welfare}} \approx \sum_{t=0}^{\infty} \beta^t \hat{w}_t^{\text{erosion}} = \underbrace{\sum_{t=0}^{\infty} \beta^t \hat{w}_t}_{\text{aggregate wage response}} - \sum_{t=0}^{+\infty} \beta^t \hat{w}_t^{\text{catch-up}}.$$

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Because the benefits of **wage catch-up** are offset by **costs of conflict**

$$\sum_{t=0}^{\infty} \beta^t \hat{w}_t^{\text{catch-up}} \approx \underbrace{\int \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \kappa_{i,t} (\mathcal{I}_{i,t} - \mathcal{I}_{i,t}^{ss}) \right] di}_{\text{aggregate costs of inflation due to conflict}}.$$

[Milgrom-Segal '02 envelope theorem applied to discrete choice set]

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- The impact of inflation on **aggregate wage** $\sum_{t=0}^{\infty} \beta^t \hat{w}_t$ **does not measure** welfare
 - Even ≈ 0 , inflation could still be costly because wage catch-up comes from costly conflicts

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- Contrast w/ purely time-dependent wage setting: $\kappa \rightarrow \infty$, $d\mathcal{W} = \sum_{t=0}^{\infty} \beta^t \hat{w}_t$
 - Special case in which aggregate real wage sufficient for welfare

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 - Special case in which aggregate real wage sufficient for welfare
- Conflict cost are **aggregate** deadweight loss, not just redistribution from workers to firms
- Extensions: (1) other aggregate shocks, (2) borrowing constraints, (3) unemployment, (4) general random menu costs

Survey Design

Motivating Evidence: Wage Growth, Inflation and Conflict

The Conflict-Cost Model

Eliciting Conflict Costs in our survey

Quantifying how conflict affects the costs of inflation

Elicit Conflict Costs with the Survey

Key parameters for welfare: conflict costs and degree of inflation indexation in default wages

Use survey to directly elicit them, then quantify how conflict impact costs of inflation

- **Elicit conflict costs** (utility costs of conflict κ & prob. of free catch-up λ)
- Elicit how default wage offer changes after inflation absent conflict (indexation γ)

Eliciting Conflict Costs

- Elicit nominal wage growth workers believe they get **via costly actions**

$$\Delta W^{\text{conflict}} \equiv \log W_{i,t}^* - \log W_{i,t-1}$$

[$W_{i,t}^*$: conflict-induced nominal wage]

What pay growth in next 12 months do you think you would get if you do your best to increase pay using any strategies at your disposal, including the common strategies listed above?

Eliciting Conflict Costs

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$[W_{i,t}^* : \text{conflict-induced nominal wage}]$

What pay growth in next 12 months do you think you would get if you do your best to increase pay using any strategies at your disposal, including the common strategies listed above?

- Elicit nominal wage growth ΔW^{indiff} where workers are **indifferent** between
 - Accepting** if offered by their employers vs **taking costly actions**
- Find x^{conflict} , the fraction of wage that workers would sacrifice to avoid conflict

$$x^{\text{conflict}} \equiv \Delta W^{\text{conflict}} - \Delta W^{\text{indiff}}$$

- Directly linked with the conflict cost κ .

Eliciting Costs of Conflict

Elicit ΔW^{indiff} via “multiple price list” standard in experimental economics

[Jack et al. '22]

	I would accept my employer's pay growth offer	I would do my best using any strategies at my disposal to increase my pay further
Employer offers you pay growth of 4%	☐	☐
Employer offers you pay growth of 3.5%	☐	☐
Employer offers you pay growth of 3%	☐	☐
Employer offers you pay growth of 2.5%	☐	☐
Employer offers you pay growth of 2%	☐	☐
Employer offers you pay growth of 1.5%	☐	☐
Employer offers you pay growth of 1%	☐	☐
Employer offers you pay growth of 0.5%	☐	☐
Employer offers you pay growth of 0%	☐	☐

Consider a respondent who reports $\Delta W^{\text{conflict}} = 4\%$. They get a list of potential employer offers $\leq 4\%$.

Eliciting Costs of Conflict

Elicit ΔW^{indiff} via “multiple price list” standard in experimental economics

[Jack et al. '22]

	I would accept my employer's pay growth offer	I would do my best using any strategies at my disposal to increase my pay further
Employer offers you pay growth of 4%	☒	☐
Employer offers you pay growth of 3.5%	☒	☐
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Employer offers you pay growth of 2%	☒	☐
Employer offers you pay growth of 1.5%	☒	☐
Employer offers you pay growth of 1%	☒	☐
Employer offers you pay growth of 0.5%	☒	☐
Employer offers you pay growth of 0%	☒	☐

Respondent never engages in conflict

Always accept employer offers & never conflict $\implies$ high conflict cost.

Eliciting Costs of Conflict

Elicit ΔW^{indiff} via “multiple price list” standard in experimental economics

[Jack et al. '22]

	I would accept my employer's pay growth offer	I would do my best using any strategies at my disposal to increase my pay further
Employer offers you pay growth of 4%	☐	☒ Respondent always engages in conflict
Employer offers you pay growth of 3.5%	☐	☒
Employer offers you pay growth of 3%	☐	☒
Employer offers you pay growth of 2.5%	☐	☒
Employer offers you pay growth of 2%	☐	☒
Employer offers you pay growth of 1.5%	☐	☒
Employer offers you pay growth of 1%	☐	☒
Employer offers you pay growth of 0.5%	☐	☒
Employer offers you pay growth of 0%	☐	☒

Never accept employer offers & always conflict $\implies$ zero conflict cost.

Eliciting Costs of Conflict

Elicit ΔW^{indiff} via “multiple price list” standard in experimental economics

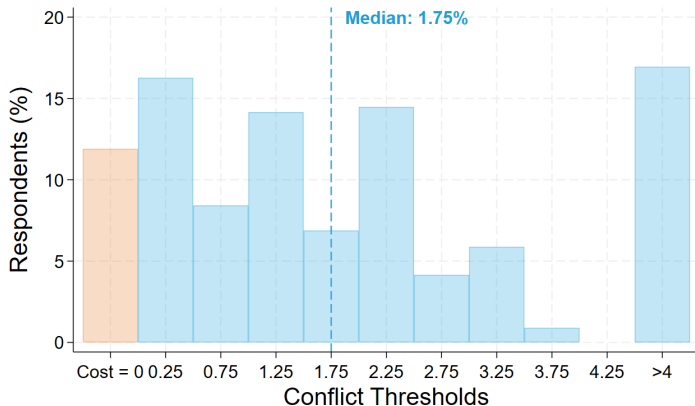
[Jack et al. '22]

	I would accept my employer's pay growth offer	I would do my best using any strategies at my disposal to increase my pay further
Employer offers you pay growth of 4%	☒	☐
Employer offers you pay growth of 3.5%	☒	☐
Employer offers you pay growth of 3%	☒	☐
Employer offers you pay growth of 2.5%	☒	☐
Employer offers you pay growth of 2%	☐	☒
Employer offers you pay growth of 1.5%	☐	☒
Employer offers you pay growth of 1%	☐	☒
Employer offers you pay growth of 0.5%	☐	☒
Employer offers you pay growth of 0%	☐	☒

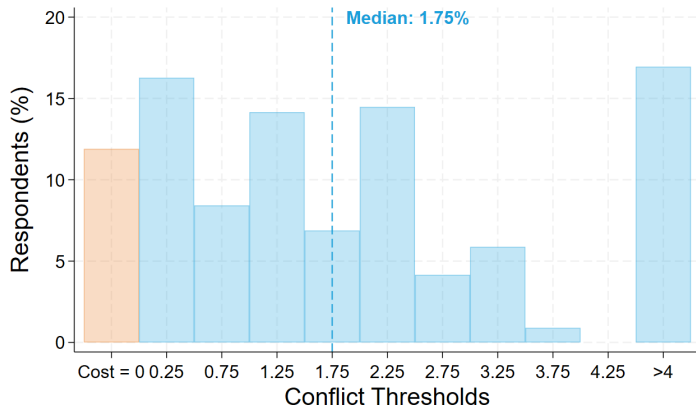
Respondent indifference wage is between 2% and 2.5% pay growth

Let ΔW^{indiff} denote lowest nominal wage growth where workers accept employers' offer, then:

$$x^{\text{conflict}} \in [\Delta W^{\text{conflict}} - \Delta W^{\text{indiff}}, \Delta W^{\text{conflict}} - \Delta W^{\text{indiff}} + 0.5].$$



- Direct evidence on **conflict costs**: $\kappa_{i,t} = 0$ with prob. λ and $\kappa_{i,t} = \kappa$ otherwise
 - $\lambda = 0.029$ (quarterly) to match share of people who would always conflict
 - Calibrate κ to median $x^{\text{conflict}} = 1.75\%$



- In paper, **validation exercises**:

- Lower conflict costs** predict **higher likelihood** of having taken **actions** in 2023
- Lower conflict costs** predict **less wage sacrificed** from not taking actions in 2023

[Here](#)
[Here](#)

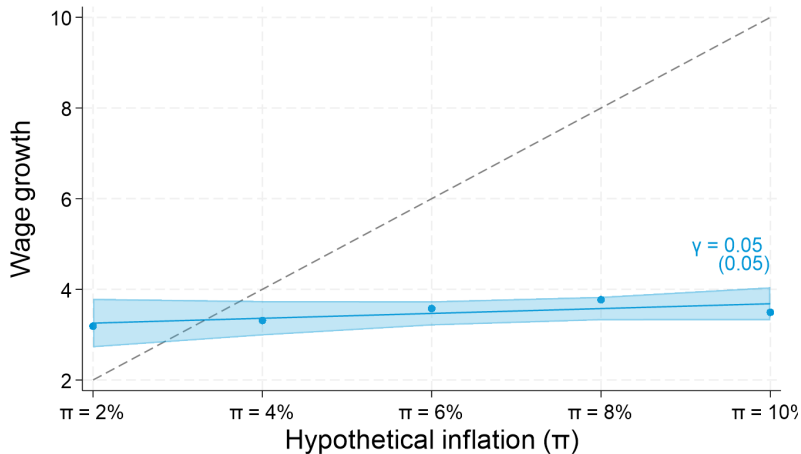
Elicit Conflict Costs with the Survey

Key parameters for welfare: conflict costs and degree of inflation indexation in default wages

Use survey to directly elicit them, then quantify how conflict impact costs of inflation

- Elicit conflict costs (utility costs of conflict κ & prob. of free catch-up λ)
- **Elicit how default wage offer changes after inflation absent conflict** (indexation γ)

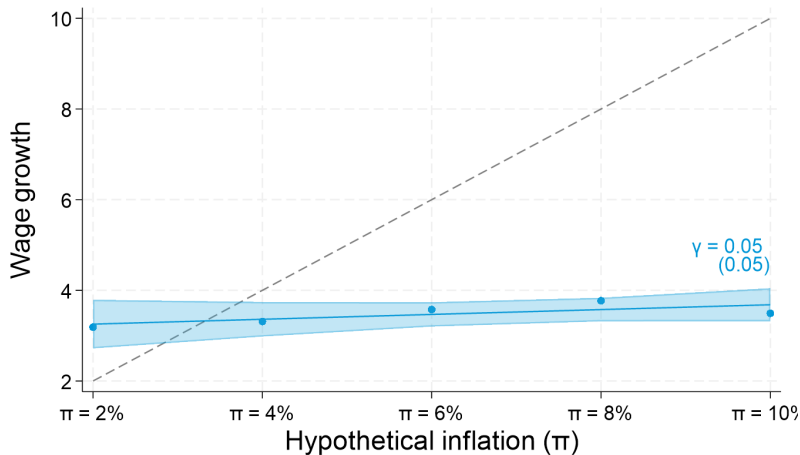
Indexation: Perceived Nominal Wage Growth without Conflict



Question:

- Consider a hypothetical situation, inflation is expected to be $x\%$ in the next 12 months.

Indexation: Perceived Nominal Wage Growth without Conflict



- Evidence on **default wages + indexation**: $W_{i,t}^{default} = W_{i,t-1} e^{\alpha + \gamma \pi_t}$
- Indexation: $\Delta \pi = 1\%$ then $\Delta \log W_{i,t}^{default} = 0.05\% \Rightarrow \gamma = 0.05$

Empirical Takeaway

1. **Significant cost of conflict** $x^{\text{conflict}} = 1.75\%$ of wages and $\lambda = 0.029$
2. **Almost no indexation**: 1 p.p. inflation $\Rightarrow \sim 0$ in employer-offered nominal wage growth

Survey Design

Motivating Evidence: Wage Growth, Inflation and Conflict

The Conflict-Cost Model

Eliciting Conflict Costs in our survey

Quantifying how conflict affects the costs of inflation

Calibration - Quarterly Frequency

	<i>Description</i>	<i>Value</i>	<i>Target</i>
β	Discount factor	0.99	Standard
κ	Conflict cost	8.14%	Own survey such that $\underline{x}^{ss} = -1.75\%$
λ	Probability of free catchup	2.87%	Own survey
g	Trend real wage growth	0.76%	ASEC-CPS 3.02% annual real wage growth
α	Default nom. wage growth at π^{ss}	0.81%	Own survey
γ	Indexation of default nominal wage	0.05	Own survey
π^{ss}	Steady state inflation	0.5%	2% annual inflation
$z_{i,t}$	Idios. shocks $z_{i,t} + \mu \sim \text{Gamma}(a, b)$	(0.14, 0.03)	$\mathbb{E}[z_{i,t}] = 0$ 48% yearly share of conflict

Quantifying the Aggregate Costs of Inflation Due to Conflict

Solve first-order responses to inflation shocks $\{d\pi_t\}_{t \geq 0}$ using **SSJ** methods

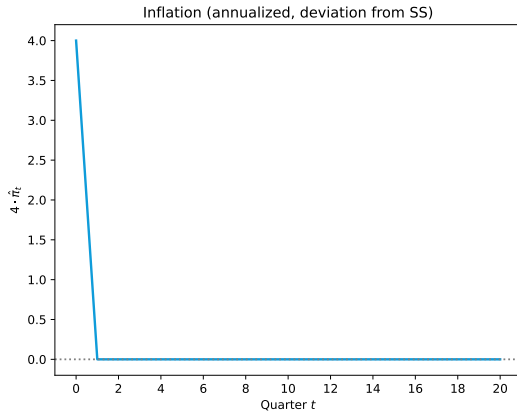
[Auclert-Bardóczy-Rognlie-Straub '21, Auclert-Rigato-Rognlie-Straub '24]

Evaluate welfare costs of inflation shocks to workers

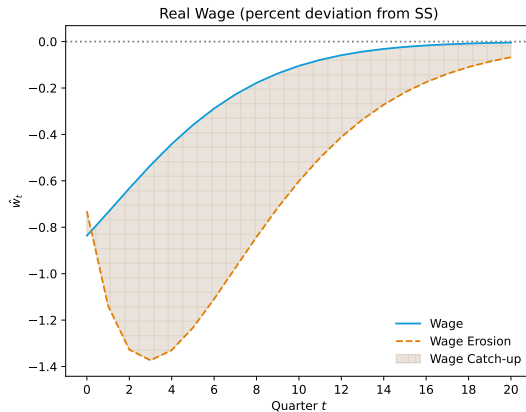
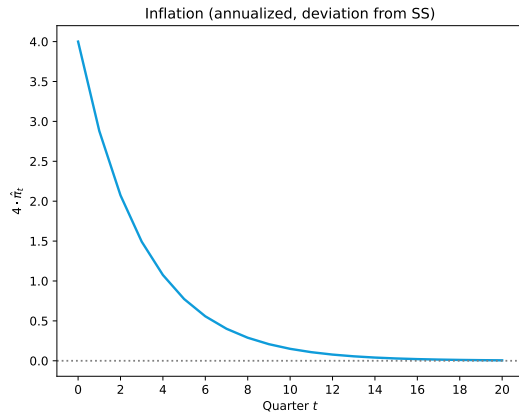
$$\hat{\mathcal{W}} = \sum_{t=0}^{\infty} \beta^t \hat{w}_t^{\text{erosion}} = \underbrace{\sum_{t=0}^{\infty} \beta^t \hat{w}_t}_{\text{aggregate wage response}} - \underbrace{\hat{\mathcal{L}}}_{\text{aggregate costs of inflation due to conflict}}.$$

1. Transitory inflation shocks: $d\pi_0 > 0$ and $d\pi_t = 0$ for $t \geq 1$
2. Persistent inflation shocks: $d\pi_t = \rho^t \varepsilon$, with $\rho = 0.72$
3. 2021-23 inflation experience (perfect foresight and observed inflation expectations)

Transitory Inflation Shocks



Persistent Inflation Shocks

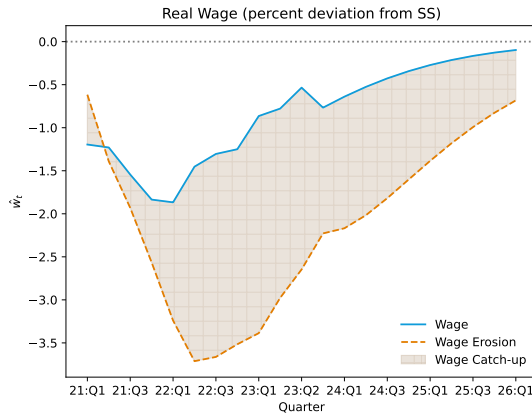
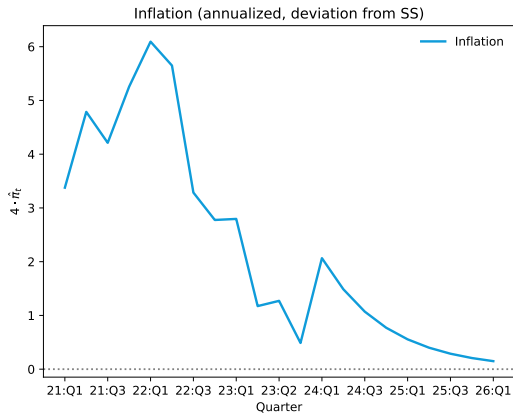


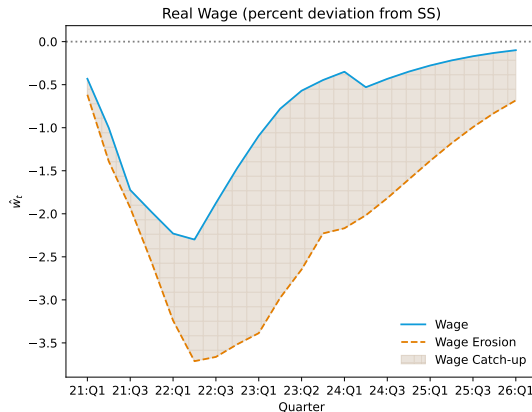
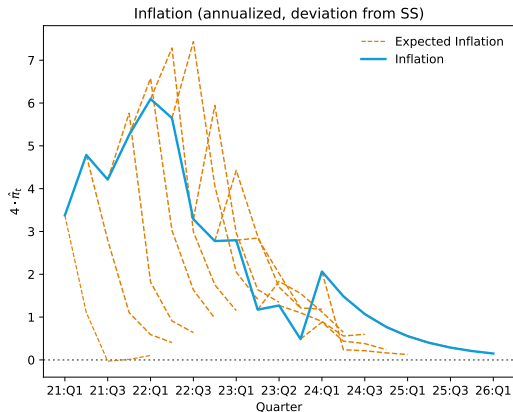
Welfare Decomposition

	Overall Welfare Change	Real Wage Response	Aggregate Costs of Inflation due to Conflict
Transitory inflation	−0.95%	−0.22%	−0.73%
Persistent inflation	−3.31%	−1.16%	−2.15%

Note: Welfare units = % of 1-year consumption

- **Costs of inflation** incorporating conflict **more than double** costs via falling real wages





Finding that aggregate costs of inflation due to conflict are significant is **robust** to

- Alternative calibration of the indexation parameter γ
- Alternative calibration of the probability of free wage catch-up λ
- Alternative calibration of the conflict cost κ
- General equilibrium determination of employment and wages

Gamma

Lambda

Kappa

GE

Conclusion

Conclusion

- Workers incur **conflict costs** to have nominal wages keep up with inflation
- Survey: workers willing to sacrifice 1.75% of wages to avoid costly conflicts
- We capture conflict cost in a menu-cost style model applied to wage setting
- Incorporating conflict costs **more than doubles** costs of inflation to workers.
- Broader agenda: a tractable approach to **introduce state-dependent wage setting**
 - In progress: firms' costs in adjusting wages away from default & downward wage rigidity

Outline

- 1 The Costs of Inflation
- 2 Overview of Search
- 3 Basic Facts
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Why Labor Search?

- Understanding unemployment is important for macroeconomics
- Why study unemployment with labor search models?

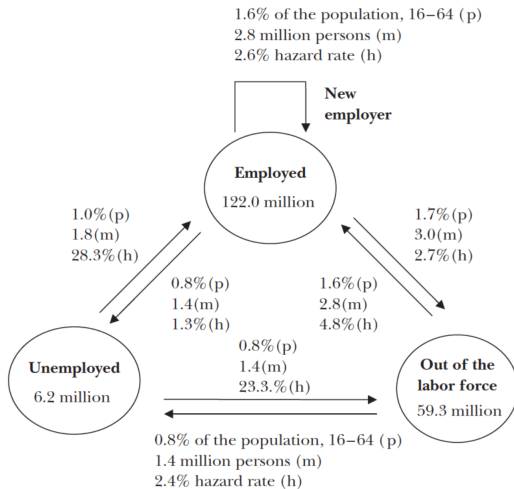
1. Labor search models generate unemployment **in equilibrium**:

- Wage rigidity without bilateral inefficiency (Hall AER 2005)
- By contrast, in New Keynesian model of labor markets:
 - Unrealistic notion of unemployment based on “unions” (E.g. Erceg, Henderson & Levin JME 2001)
 - Wage rigidity violates bilateral efficiency (Barro JME 1977)

Why Labor Search?

- Understanding unemployment is important for macroeconomics
 - Why study unemployment with labor search models?
-
- 2. Overall numbers like unemployment rates mask substantial churn
 - Constant hiring and firing, creation of new jobs and destruction of old
 - Prior to recession, ~ 10 percent of U.S. workers separated from employers per quarter
 - Motivates focus on flows rather than stocks.
 - Labor search is a theory of labor market flows

Labor Market Flows



Search: Macro Versus Micro

- Main focus today is on macro search:
 - General equilibrium
 - Focus on vacancy posting by firms
- Many interesting questions in micro search:
 - Often in partial equilibrium: takes vacancy posting as given
 - Models individuals' job search
 - Study details of matching function
 - Relates wage dispersion and search frictions to inequality
- Obviously (1) and (2) aren't fully separate

The Shimer Puzzle

- Canonical Diamond-Mortensen-Pissarides model does not generate unemployment volatility
- Basic problem:
 - Fall in labor demand leads to fall in wages
 - Profits from vacancy creation remain stable
 - Vacancy creation falls very little in response to fall in labor demand
- Lots of proposed solutions
 - Including but not exclusively wage rigidity

Focus on Hiring Margin Today

- Canonical DMP model features **exogenous separations**:
 - Unemployment determined by hiring fluctuations
 - Roughly consistent with the data (to be discussed)
- Models emphasizing separations
 - Mortensen & Pissarides (ReStud 1994) has endogenous job destruction
 - Coles & Kelishomi (2018) has separation shocks
- Focus on hiring means studying vacancy creation

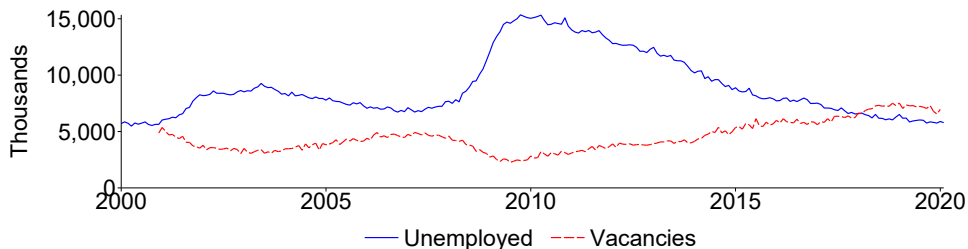
The Plan for the Rest of Today

- Basic Facts
- Models—Shimer puzzle and proposed resolutions
- Empirics of wage rigidity
- Scarring from job loss and the link to hysteresis

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Prima Facie Evidence of Matching Frictions



- Median unemployment duration: 5-25 weeks.
- Average vacancy duration: 1 month.

Large Gross Flows

Share of working-age population making labor force status transitions

	E_t	U_t	N_t	All
E_{t-1}	59.15	0.77	1.65	61.58
U_{t-1}	0.88	1.95	0.83	3.66
N_{t-1}	1.54	0.84	32.40	34.77
All	61.57	3.56	34.88	100.00

- Flows from N to E *larger* than flows from U to E .
- Transitions into and out of labor force historically viewed as acyclical.
- Transition *hazard* much larger for U than N .

“Ins and Outs” of Unemployment

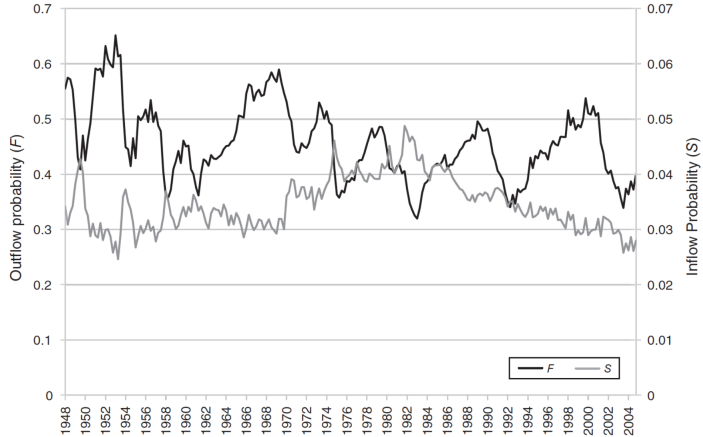
- Is the employment exit / separation / job destruction (“ins”) or employment entry / job finding / job creation margin (“outs”) more important?
- Conventional wisdom from 80s and 90s: Ins more important.
- Shimer (RED 2012) argues 3/4 of fluctuations in the unemployment rate — 90% since 1987 – are due to the job finding margin
- Leads to focus on job finding rate in recent literature.

	1	2	3	4	5
	all workers 1948–2010	all workers 1967–2010	all workers 1976–2010	all workers 1987–2010	men 25–54 1976–2010
$\frac{\bar{x}}{\bar{x} + \bar{f}}$	0.77	0.83	0.85	0.90	0.69
$\frac{x_t}{x_t + f}$	0.24	0.17	0.15	0.10	0.32

“Ins and Outs” of Unemployment

- While Hall (ReStat 2005) agrees with Shimer, Fujita and Ramey (IER 2009) challenges
- Elsby, Michaels & Solon (AEJM 2009) modify:
 - Job finding rate dominates.
 - But inflows matter in severe recessions.
 - Focusing on aggregate inflow rate masks fluctuations in reasons for inflows
 - $\text{Separations} = \text{Layoffs} + \text{Quits}$
 - Layoffs strongly countercyclical but quits procyclical

“Ins and Outs” of Unemployment



Source: Elsby, Michaels & Shapiro (2009)

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Ingredients

- Agents employed (mass e) or unemployed (mass u), $e + u = 1$.
- Mass v of vacancies posted at cost c per vacancy.
- Matching function $m(u, v) = Mu^{1-\eta}v^\eta$.
- Market tightness: $\theta = v/u$.
- Exogenous separation probability s .
- Agent produces p if employed or z if unemployed.
- Discount future with factor δ .

Model Equations

■ Value functions:

$$\text{Unemployment: } U = z + \delta \{ f(\theta) E[W'] + [1 - f(\theta)] E[U'] \}. \quad (1)$$

$$\text{Employment: } W = w + \delta \{ [1 - s] E[W'] + sE[U'] \}. \quad (2)$$

$$\text{Job: } J = p - w + \delta \{ [1 - s] E[J'] + sE[V'] \}, \quad (3)$$

$$\text{Vacancy: } V = -c + \delta \{ q(\theta) E[J'] + [1 - q(\theta)] E[V'] \}. \quad (4)$$

■ Free entry condition:

$$V = 0. \quad (5)$$

■ Endogenous transition probabilities:

$$\text{Job-finding: } f(\theta) = \frac{m(u, v)}{u} = M\theta^\eta. \quad (6)$$

$$\text{Vacancy-filling: } q(\theta) = \frac{m(u, v)}{v} = M\theta^{\eta-1} = \frac{f(\theta)}{\theta}. \quad (7)$$

Next Steps

1. Treat w as free parameter and solve model in steady state as $\delta \rightarrow 1$.
2. Comparative statics of this model.
3. Comparative statics with Nash bargaining.
4. Chodorow-Reich and Karabarbounis critique.

Solution

(3) and (5)

$$J = (p - w) / s,$$

(4) and (5)

$$q = c / J$$

Combine:

$$= \frac{cs}{p - w},$$

(7)

$$q = M\theta^{\eta-1},$$

Combine:

$$\theta = \left(\frac{cs}{M(p - w)} \right)^{\frac{1}{\eta-1}},$$

(1) and (2)

$$W - U = (w - z) + (1 - s - f)(W - U)$$

$$= \frac{w - z}{s + f},$$

Match surplus:

$$S = J + W - U.$$

Unemployment Volatility

- We solved the model assuming time-invariant parameters.
- To consider volatility, we will take a short-cut and compare across steady-states.
- This turns out to be okay because labor market converges very quickly to steady state in this model.
 - Intuition: all endogenous variables “jump” other than unemployment
- Not okay in richer models with more dynamics such as due to capital or other slow moving state variables.
- Formally, comparative static of θ w.r.t. p allowing for wage to change endogenously in response to p .

Comparative Static

$$\theta = \left(\frac{cs}{M(p-w)} \right)^{\frac{1}{\eta-1}},$$

$$\frac{d \ln \theta}{d \ln p} = \left(\frac{1}{1-\eta} \right) \frac{d \ln(p-w)}{d \ln p} = \left(\frac{1}{1-\eta} \right) \left(\frac{p}{p-w} \right) \left(1 - \frac{dw}{dp} \right).$$

- If wages increase one-for-one with p , $dw/dp = 1$, no extra incentive for firms to hire and no change in unemployment.
- If wages are sticky, $dw/dp = 0$, then job-finding increases by $p/(p-w)$. Ljungvist and Sargent call this *fundamental surplus*.
- Closer is w to p , larger is percent increase in firm profits if p increases.

Comparative Static

$$\theta = \left(\frac{cs}{M(p-w)} \right)^{\frac{1}{\eta-1}},$$
$$\frac{d \ln \theta}{d \ln p} = \left(\frac{1}{1-\eta} \right) \frac{d \ln(p-w)}{d \ln p} = \left(\frac{1}{1-\eta} \right) \left(\frac{p}{p-w} \right) \left(1 - \frac{dw}{dp} \right).$$

Mechanisms generating unemployment volatility:

1. Wage rigidity— dw/dp small (Hall 2005, Hall & Milgrom 2008, Gertler & Trigari 2009)
2. Small fundamental surplus— $p/(p-w)$ small (Hagedorn & Manovskii 2008)

These two mechanisms are complementary

Nash Wage Equation

- Endogenize w with **standard** Nash assumption:

$$J = (1 - \beta)S, (W - U) = \beta S.$$

- With lots of messy algebra, can solve for w :

$$w = z + (1 - \beta)(p - z) + \beta \theta c.$$

- Period wage is worker's flow opportunity cost z , plus share of surplus $(1 - \beta)(p - z)$, plus labor market tightness term.
- Nash wage is procyclical (comoves with p), because flow surplus is procyclical and because labor market tightness comoves with p .
- Intuition: worker's threat point in bargain is to return to unemployment pool. Value of unemployment depends on flow value z and likelihood of finding another job soon, which is increasing in labor market tightness.

Shimer (AER, 2005) puzzle

- With Nash wages we have

$$\frac{d \ln \theta}{d \ln p} = \left(\frac{1}{1 - \eta} \right) \frac{d \ln(p - w)}{d \ln p} = \left(\frac{1}{1 - \eta} \right) \left(\frac{p}{p - w} \right) \left(1 - \frac{dw}{dp} \right)$$

$$w = z + (1 - \beta)(p - z) + \beta \theta c.$$

- Shimer sets $p = 1, z = 0.4, 1 - \beta = 0.72$.

- Therefore

$$\left(\frac{p}{p - w} \right) \quad \text{small}$$

$$\frac{dw}{dp} \quad \text{near } 1$$

$$\frac{d \ln \theta}{d \ln p} \quad \text{small} \approx 1/10 \text{ of empirical value}$$

- This is the Shimer puzzle!

Hagedorn & Manovskii (AER, 2008) calibration

- With Nash wages we have

$$\frac{d \ln \theta}{d \ln p} = \left(\frac{1}{1 - \eta} \right) \frac{d \ln(p - w)}{d \ln p} = \left(\frac{1}{1 - \eta} \right) \left(\frac{p}{p - w} \right) \left(1 - \frac{dw}{dp} \right)$$

$$w = z + (1 - \beta)(p - z) + \beta \theta c.$$

- Hagedorn & Manovskii set $p = 1, z = 0.95$

- Therefore

$$\left(\frac{p}{p - w} \right) \quad \text{large}$$

$$\frac{d \ln \theta}{d \ln p} \quad \text{large} \approx \text{of empirical value}$$

- This is the Hagedorn & Manovskii resolution: example of small *fundamental surplus*.

What About Wage Rigidity?

- But how much wage rigidity is there? I.e. how large is dw/dp ?
- What sort of wage rigidity matters?

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What Kind of Wage Rigidity Matters?

- Pissarides (ECMA 2009) points out that the relevant wage is for new hires
 - Not continuing/incumbent/ongoing workers
- Intuition:
 - What matters for vacancy is **present value of profits** at start of job
 - Because employment is a long term contract
 - Depends on present value of wages at start of job
 - Closely tied to wage for new hires

Revisiting Model Outside Steady State

- Value of job w/ free entry

$$J_t = p_t - w_t + \delta \{ [1 - s] E_t [J_{t+1}] + s E [V_{t+1}] \}$$

- Value of vacancy w/ free entry

$$V_t = -c + \delta \{ q(\theta_t) E_t [J_{t+1}] + [1 - q(\theta_t)] E_t [V_{t+1}] \}$$

- Free entry

$$V_{t+1} = 0$$

- Implies

$$\frac{c}{\delta q(\theta_t)} = E_t [J_{t+1}]$$

$$E_t J_{t+1} = E_t \sum_{j=0}^{\infty} [\delta (1 - s)]^j (p_{t+j+1} - w_{t+j+1}).$$

Revisiting Model Outside Steady State

- Assume (for simplicity): p_t is a driftless random walk

$$\begin{aligned} \frac{c}{\delta q(\theta_t)} &= E_t \sum_{j=0}^{\infty} [\delta(1-s)]^j (p_{t+1} - w_{t+1}) \\ &= \left(\frac{p_t - \tilde{w}_t}{1 - \delta(1-s)} \right) \quad \tilde{w}_t = \overbrace{[1 - \delta(1-s)] E_t \sum_{j=0}^{\infty} [\delta(1-s)]^j w_{t+j+1}}^{\text{Flow expected present value of wages}} \end{aligned}$$

- Log differentiate both sides:

$$\frac{d \log \theta_t}{d \log p_t} = \left(\frac{1}{1 - \eta} \right) \frac{p_t}{p_t - \tilde{w}_t} \left(1 - \frac{d \tilde{w}_t}{d p_t} \right)$$

(This result is a generalization of Pissarides 2009)

Present Value of Wages is Allocative

$$\frac{d \log \theta_t}{d \log p_t} = \left(\frac{1}{1 - \eta} \right) \frac{p_t}{p_t - \tilde{w}_t} \left(1 - \frac{d \tilde{w}_t}{d p_t} \right)$$

$$\tilde{w}_t = [1 - \delta(1 - s)] E_t \sum_{j=0}^{\infty} [\delta(1 - s)]^j w_{t+j+1}$$

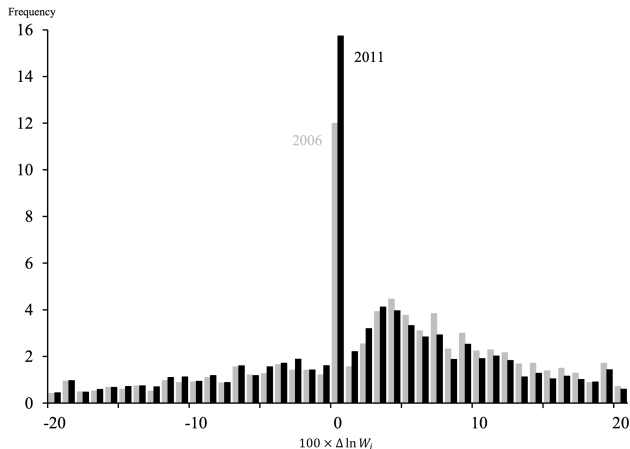
- Rigidity of **flow expected present value of wages** $\tilde{w}_t$ is what matters
- Present value is closely tied to wage at start of match w_{t+1}
- $\tilde{w}_t$ and w_{t+1} are perfectly correlated if w_{t+j} varies little after start of match
 - Consistent with data on wage rigidity for continuing workers
- Therefore wage for new hire is (approximately) allocative
- Note: argument relies strongly on exogenous separations (i.e. baseline DMP)

Wage Rigidity: New Hires Versus Continuing Workers

- What does previous work (before Pissarides 2009) do?
 - Calibrates to wage rigidity for **continuing workers**
 - I.e. incumbent workers in ongoing employment
 - Ample evidence of wage rigidity for continuing workers

Downward Wage Rigidity for Continuing Workers

Figure 1. Distribution of 12-month log wage changes in 2006 and 2011.



Source: Current Population Survey and authors' calculations.

Wage Rigidity: New Hires Versus Continuing Workers

- But maybe wages are more flexible for new hires than continuing workers
 - Maybe firms insure continuing workers against income shocks but not new hires (Harris & Holmstrom 1982; Beaudry & DiNardo 1991)
 - Maybe continuing workers have a reference point of their own past wage
 - Whereas new hires have no reference point
- What do the data say?

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Pissarides' Review of Literature

- Pissarides reviews wage cyclicalities for job switchers, i.e. regressions like

$$\Delta \log w_{it} = \alpha + \psi \Delta u_t + \varepsilon_{it}$$

where:

- u_t is the unemployment rate
 - w_{it} is the wage for **worker switching jobs** i.e. a new hire
 - Typically run on panel data (e.g. NLSY, PSID or CPS for US)
- Finds that wages for new hires are very procyclical
 - Estimates of ψ_u imply calibration of $d\tilde{w}_t/dp_t \approx 1$
 - Concludes wage rigidity cannot be mechanism for unemployment fluctuations

Pissarides' Review of Literature

TABLE II
ESTIMATES OF THE CYCLICALITY OF HOURLY WAGES, UNITED STATES^a

Author	Data		Coefficient on $-\Delta u \times 100$
Bils (1985)	NLSY 1966–1980	All (whites/nonwh.)	1.6/1.8
		Stayers	0.6/0.4
		Changers	3.0/4.0
Shin (1994)	NLSY 1966–1981	All (whites/nonwh.)	1.7/1.4
		Stayers	1.2/0.2
		Changers	2.7/3.8
Barlevi (2001)	PSID, 1968–1993 NLSY, 1979–1993	Changers	2.59
		Changers	3.00
Beaudry and DiNardo (1991)	PSID 1976–1984	All, cont. u	0.7
		All, initial u	0.6
		All, min. u	2.9
	CPS 1979, 1983	All, cont. u	0.0
		All, initial u	0.0
		All, min. u	3.1
Grant (2003)	NLSY 1966–1981	All, cont. u	2.37
		All, initial u	0.60
		All, min. u	2.29
Solon, Barsky, and Parker (1994)	PSID 1968–1987	All men	1.40
		All women	0.53
		Stayers, men	1.24
Devereux (2001)	PSID 1970–1991	All	1.16
		Stayers	0.81
		Single job holders	0.54
Shin and Solon (2006)	NLSY 1979–1993	All	1.37
		Stayers	1.17
		Single job holders	1.13

^aThe dependent variable is the annual change in the log of hourly earnings, obtained from the estimated coefficients on annual time dummies in individual wage regressions. Results are for men, unless otherwise stated. Unemployment is national unemployment in percent, except for Barlevi's study, which uses state unemployment. In Beaudry and DiNardo's and Grant's studies the results shown are from regressions with three independent unemployment variables: (i) contemporaneous unemployment; (ii) unemployment at start of job; (iii) lowest unemployment since start of job. Acronyms used: National Longitudinal Surveys of Youths (NLSY); Current Population Survey (CPS).

Source: Pissarides (2009)

But What About Composition?

But What About Composition?

- Gertler & Trigari (JPE 2009) critique:
 - Cyclicalities could come from heterogeneous match quality
- Example:
 - Banker who switches to barista job in a recession has procyclical earnings
 - But this cyclicalities is irrelevant from perspective of bank or coffee shop
 - Bank or coffee shop are making vacancy creation decision
 - Sometimes known as “composition bias”

Gertler, Trigari & Huckfeldt's Solution

- Gertler, Trigari & Huckfeldt (ReStud 2020) argue:
 - Composition probably very important for job switchers (“cyclical upgrading”)
 - Composition probably less important for workers hired from unemployment (Unemployed likely to be less selective)
- Gertler et al run the regression

$$\Delta \log w_{it} = \alpha + \psi_u \Delta u_t \times I(ENE_{it}) \\ + \psi_e \Delta u_t \times I(EE_{it}) + \varepsilon_{it}$$

where:

- u_t is the unemployment rate
- w_{it} is the wage for new hires
- $I(ENE_{it}), I(EE_{it})$ indicators for whether job switcher or entering from unemployment
- Estimated using SIPP (high frequency panel survey data since 1990)

Results Suggest Wage Rigidity for New Hires

	First differences		Fixed-effects	
	(1)	(2)	(3)	(4)
UR	-0.425*** (0.0967)	-0.420*** (0.0966)	-0.145** (0.0609)	-0.145** (0.0609)
UR · $\mathbb{I}(EE)$	-1.878*** (0.6800)	-1.674*** (0.6223)	-1.982*** (0.5046)	-1.941*** (0.4730)
UR · $\mathbb{I}(ENE)$	-0.439 (0.6639)	-0.549 (0.7346)	-0.337 (0.5387)	0.054 (0.5956)
$\mathbb{I}(EE)$	0.045*** (0.0048)	0.038*** (0.0046)	0.004* (0.0023)	0.001 (0.0022)
$\mathbb{I}(ENE)$	-0.047*** (0.0066)	-0.065*** (0.0074)	-0.030*** (0.0029)	-0.034*** (0.0034)
$P(b_{nu}^{EE} = b_{nu}^{ENE})$	0.126	0.238	0.023	0.008
Unemp. spell for ENE	0+	1+	0+	1+
No. observations	318,768	318,768	375,649	375,649
No. individuals	56,881	56,881	56,881	56,881
No. EE new hires	8,714	10,124	9,857	10,124
No. ENE new hires	5,331	3,921	6,437	4,858

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Dependent variable: log hourly real wage. Controls for education, union coverage, marital status, a quadratic in tenure, and a linear time trend. Robust standard errors in parenthesis, clustered by individual.

Results Suggest Wage Rigidity for New Hires

- Results:
 - Wage for job switchers is strongly cyclical
 - Wage for new hires from unemployment is weakly cyclical
- Suggests that composition-adjusted wage for new hires are weakly cyclical
 - Contra Pissarides (2009)
- Lingering concern:
 - Is the wage for new hires from unemployment composition-free?

Hazell & Taska (2025)

New dataset:

- Burning Glass online vacancy data with job titles, establishment identifiers
- Subset (17%) of online vacancies include posted wage
- Data from 2010-21, so includes variation from the pandemic

Main regression:

$$\Delta \log w_{jst} = \alpha + \psi \Delta u_{st} + \varepsilon_{jst}$$

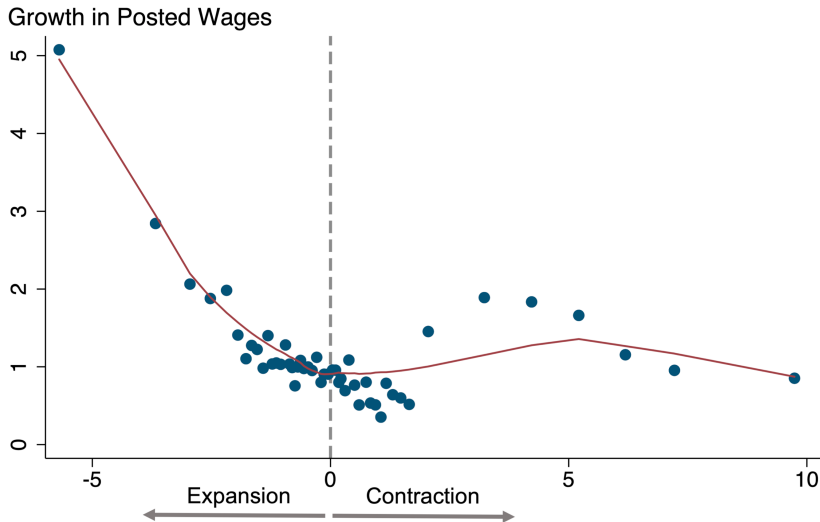
where:

- w_{jst} is the wage posted by job j in state s in quarter t
- Job = job title within establishment (e.g. barista at Starbucks in Manhattan)

Key advantage:

- Difference at the job level $\rightarrow$ directly correct for job composition

Wage for New Hires: Rigid Downward, Flexible Upward

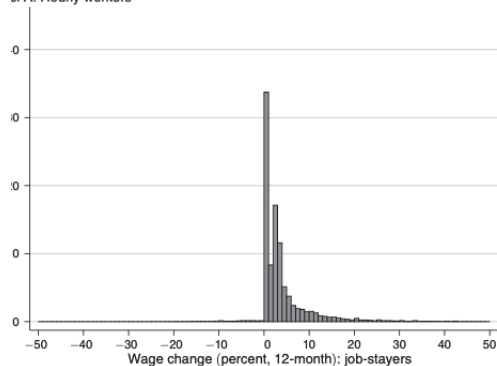


Grigsby, Hurst, Yildirmaz (AER, 2021)

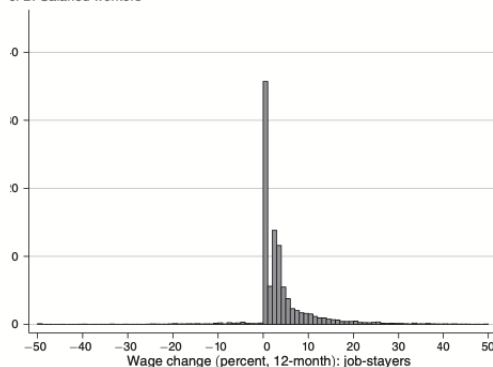
- Administrative data from payroll processor ADP over 2008-16.
- Base wages of job-stayers are downwardly rigid.
- New hires wages no more cyclical than job-stayers (composition adjusted).

Downward Rigidity in Base Wages

al A. Hourly workers



al B. Salaried workers



Job Switchers Wages More Cyclical...

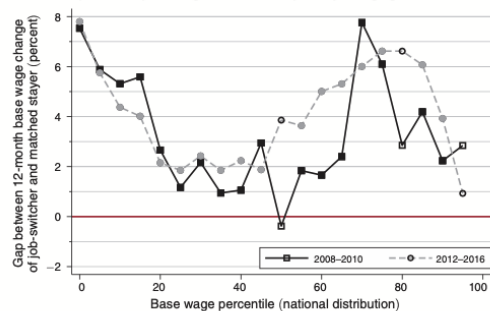
TABLE 6—CYCLICALITY OF NEW HIRE WAGES

	(1)	(2)	(3)	(4)
ΔU_{st} (percentage points)	−0.33 (0.03)	−0.22 (0.02)	−0.35 (0.03)	−0.35 (0.03)
$\Delta U_{st} \times Switcher$	−0.77 (0.24)	−0.80 (0.21)	−0.69 (0.12)	−0.18 (0.17)
State fixed effects	Yes	Yes	Yes	Yes
Demographic and wage percentile controls	Yes	Yes	Yes	Yes
Firm fixed effects	No	Yes	Yes	Yes
Interacted controls	No	No	Yes	Yes
Tenure control	No	No	No	Yes
Observations (thousands)	12,514	12,514	12,514	12,514

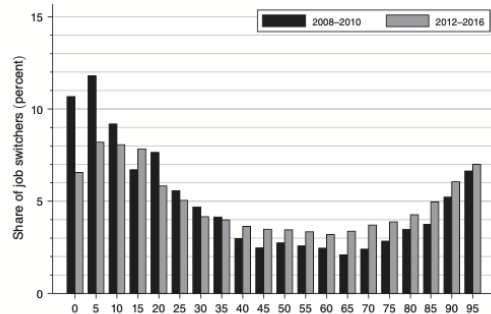
Notes: Table reports cyclicalities of wage growth for job-stayers and job-changers, estimated from equation (3). The dependent variable in columns 1–5 is the percentage base wage growth between month t and month $t + 12$, while the independent variable is the change, in percentage points, in state unemployment rates between t and $t + 12$. All columns include controls for demographics, namely fixed effects for 5-year age bins and worker sex, and the worker's percentile in the national base wage distribution as of month t . Columns 2–4 additionally include firm fixed effects, while columns 3–4 include fully interacted controls for firm, demographics, and wage percentile. Column 4 additionally controls for worker tenure in period t . Standard errors, clustered at the month level, are reported in parentheses.

But Due to Composition

Panel A. Gap between job-changer and matched job-stayer wage growth



Panel B. Share of job-changers by initial wage percentile



If Wages Are Rigid, Then Why?

- Hall (AER 2005): any wage remaining in bargaining set satisfies bilateral efficiency and is valid outcome. So assume rigid wages, for example due to social norms.
- Internal equity between continuing workers and new hires (e.g. Gertler & Trigari 2009)
- Problem: very little systematic evidence on wage determination process. Bargaining? Wage posting? Other?

If Wages Are Flexible, Then What?

- Countercyclical r :
 - Hiring cost = $\frac{p-w}{r+s} \rightarrow$ determines vacancy creation
 - Suppose that $p - w$ is acyclical due to wage procyclicality
 - Higher discount rate reduces job value
 - (Hall, AER 2017; Nadeu and Wasmer, AEJ:Macro 2013; Kehoe, Lopez, Midrigan, Pastorino, Restud 2021)

Discount Rates and Unemployment Fluctuations

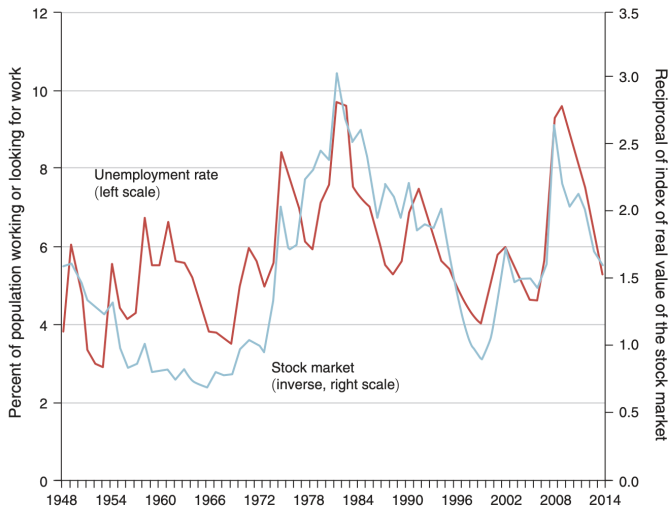


FIGURE 2. UNEMPLOYMENT AND THE REAL, DETRENDED VALUE OF THE S&P STOCK MARKET INDEX, 1948–2015

Outline

- 1 The Costs of Inflation
- 2 Overview of Search
- 3 Basic Facts
- 4 Basic Search Model
 - Importance of Wage for New Hires
- 5 Empirics of Wage for New Hires
- 6 Costs of Job Loss**
- 7 Conclusion

Cost of Job Loss

- Switching gear to talk about cost of job loss
- In the data: costs of job loss are very large and persistent
 - Workers who are fired lose **~2 years** of pre-displacement earnings
 - Robust fact across many studies
- Why is this important?
 1. Huge cost of unemployment fluctuations
 2. Persistence implies long run effects of business cycles i.e. hysteresis
 3. Benchmark DMP model cannot account for costs of job loss in data → need for new models?

Jacobson, LaLonde, and Sullivan (1993): Setup

- Administrative data from Pennsylvania in 1980s.
- Examine earnings of displaced high-tenure (6+ years) workers
 - Focus on “mass layoff” = firm contracts by at least 30% in previous two years
- Event study design:

$$y_{it} = \alpha_i + \gamma_t + \beta x_{it} + \sum_{k \geq -m} \delta_k D_{it}^k + \varepsilon_{it}$$

- k is periods since layoff starting m periods ahead
 - D_{it}^k is indicator for k periods since displacement
 - x_{it} is time varying observables
 - Worker FE control for permanent observables
 - Also add worker specific linear trends
- What is the treatment group? What is the control group?

JLS (1993): Concerns About Mass Layoff Design

1. Mass layoffs could have quit or been discharged for cause.
 - Usually argue “most” discharged exogenously.
 2. Mass layoffs include firms with sharp employment contractions that do not exit, which may lay off worst workers first.
 3. Similarly, includes firms that were already doing badly, and best workers may have already quit.
 4. This is an event study!
 - AFAIK has not been replicated with “modern methods”
- Concerns 2 and 3 limited by worker FE, but do not address time-varying unobservables.

JLS (1993): Main Result

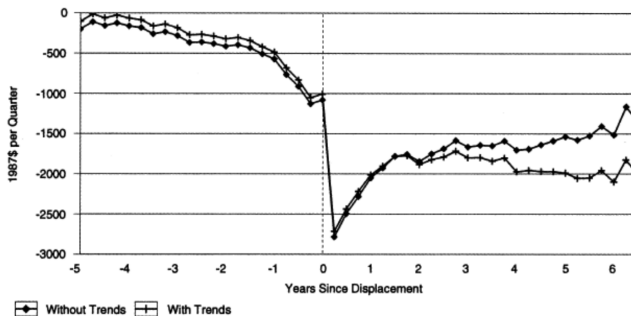
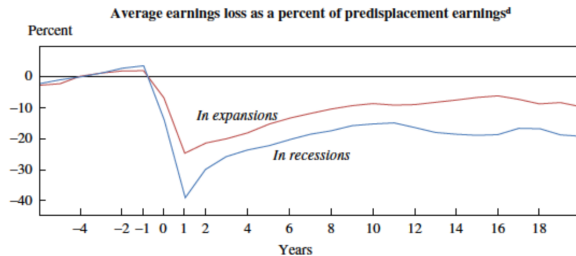
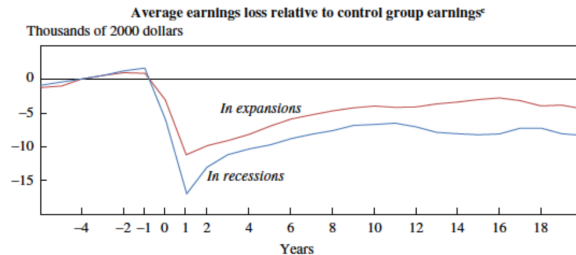


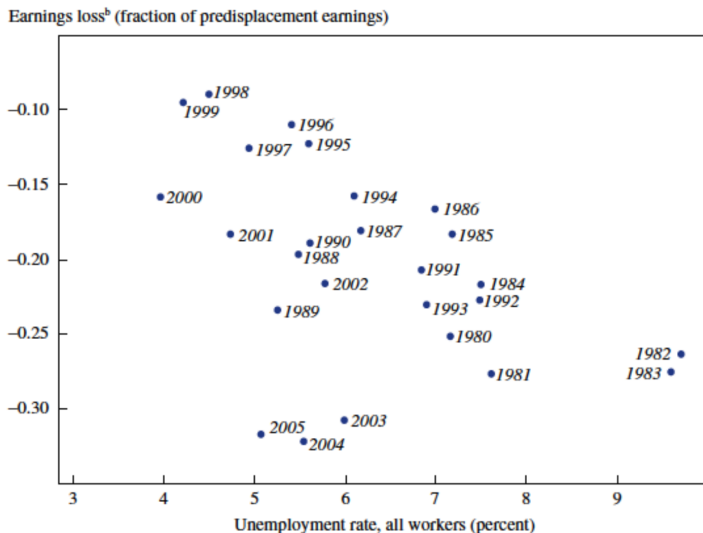
FIGURE 2. EARNINGS LOSSES FOR SEPARATORS IN MASS-LAYOFF SAMPLE

- Strength and persistence depend primarily on local labor market condition and former industries, but large across sectors and even for workers who find jobs at similar firms.

Davis and Von Wachter (2012): Recession vs. Boom



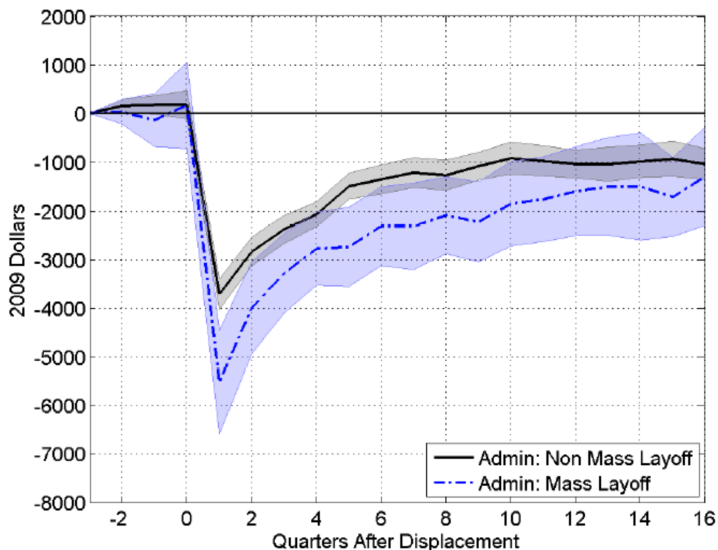
Davis and Von Wachter (2012): Loss vs. Unemp



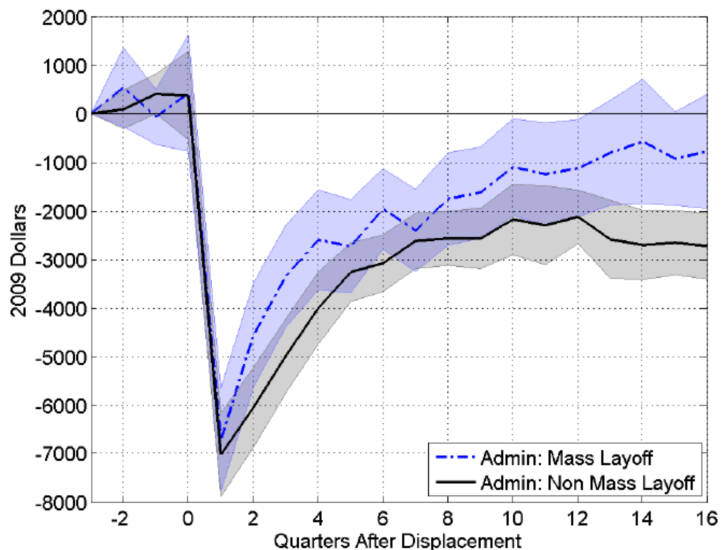
Flaaen, Shapiro, and Sorkin (2017): Survey and Admin

- Merge SIPP to LEHD.
 - LEHD is administrative data, can look at mass layoffs.
 - Worker surveys like SIPP ask reason for separation.
- Use survey reason for layoff to address concern that mass layoffs could have quit or been discharged for cause
 - I.e. addresses identification concerns with JLS

FSS (2017): Admin Data Mass Layoff (No Survey Used)



Similar Results with Survey Reason for Layoff



Scarring in DMP: Davis and Von Wachter (2012)

- DMP models have little scarring:
 - Homogenous job finding rate, separation rate, wage
 - No “wage premium” of good match to lose
 - Unemployment spells brief, followed by return to employment and common wage
- Davis and Von Wachter (2012) show job loss is inconsequential in DMP:
 - Calculate mean decline in PDV of income
 - Standard DMP: 0.2%
 - Hagedorn and Manovskii: .04%
 - Data: 15-25%

Open Questions

No settled explanations for scarring effect

Some possibilities:

- Jarosch (2015): frictional job ladder model with “slippery bottom rungs”
- Huckfeldt (2021): laid off workers switch to lower skill occupations

Open (and important) question:

- What are the consequences for aggregate hysteresis?
- Fall in wage for individual workers after job loss are very persistent
- Implies very persistent fall in economy wide productivity after recessions?

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Conclusion

- New costs of inflation from the labor market
- Wage rigidity and unemployment fluctuations
- Our final two lectures:
 - Micro evidence on consumption
 - Automation